MSG No. 70.531 $Fd'd'd'$ [Type III, orthorhombic]

Table 1: Wyckoff site: `8a`, site symmetry: `222`

No.	position	mapping
1	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right]$	`[1,10,19,28]`
2	$\left[\frac{1}{8}, \frac{5}{8}, \frac{5}{8}\right]$	`[2,9,20,27]`
3	$\left[\frac{5}{8}, \frac{1}{8}, \frac{5}{8}\right]$	`[3,12,17,26]`
4	$\left[\frac{5}{8}, \frac{5}{8}, \frac{1}{8}\right]$	`[4,11,18,25]`
5	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	`[5,14,23,32]`
6	$\left[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}\right]$	`[6,13,24,31]`
7	$\left[\frac{3}{8}, \frac{7}{8}, \frac{3}{8}\right]$	`[7,16,21,30]`
8	$\left[\frac{3}{8}, \frac{3}{8}, \frac{7}{8}\right]$	`[8,15,22,29]`

Table 2: Wyckoff site: `8b`, site symmetry: `222`

No.	position	mapping
1	$\left[\frac{1}{8}, \frac{1}{8}, \frac{5}{8}\right]$	`[1,10,19,28]`
2	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	`[2,9,20,27]`
3	$\left[\frac{5}{8}, \frac{1}{8}, \frac{1}{8}\right]$	`[3,12,17,26]`
4	$\left[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}\right]$	`[4,11,18,25]`
5	$\left[\frac{7}{8}, \frac{7}{8}, \frac{3}{8}\right]$	`[5,14,23,32]`
6	$\left[\frac{7}{8}, \frac{3}{8}, \frac{7}{8}\right]$	`[6,13,24,31]`
7	$\left[\frac{3}{8}, \frac{7}{8}, \frac{7}{8}\right]$	`[7,16,21,30]`
8	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	`[8,15,22,29]`

Table 3: Wyckoff site: `16c`, site symmetry: `-1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5]`
2	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	`[2,14]`
3	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	`[3,23]`
4	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[4,32]`
5	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	`[6,10]`
6	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	`[7,19]`
7	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[8,28]`
8	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[9,13]`
9	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[11,31]`
10	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[12,24]`
11	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[15,27]`
12	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[16,20]`
13	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[17,21]`
14	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[18,30]`
15	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[22,26]`
16	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[25,29]`

Table 4: Wyckoff site: `16d`, site symmetry: `-1'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,5]`
2	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	`[2,14]`
3	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[3,23]`
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[4,32]`
5	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[6,10]`
6	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[7,19]`
7	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[8,28]`
8	$\left[\frac{1}{2}, 0, 0\right]$	`[9,13]`
9	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[11,31]`
10	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[12,24]`
11	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[15,27]`
12	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[16,20]`
13	$\left[0, \frac{1}{2}, 0\right]$	`[17,21]`
14	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[18,30]`
15	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[22,26]`
16	$\left[0, 0, \frac{1}{2}\right]$	`[25,29]`

Table 5: Wyckoff site: `16e`, site symmetry: `2..`

No.	position	mapping
1	$\left[x, \frac{1}{8}, \frac{1}{8}\right]$	`[1,10]`
2	$\left[x, \frac{5}{8}, \frac{5}{8}\right]$	`[2,9]`
3	$\left[\frac{3}{4} - x, \frac{1}{8}, \frac{5}{8}\right]$	`[3,12]`
4	$\left[\frac{3}{4} - x, \frac{5}{8}, \frac{1}{8}\right]$	`[4,11]`
5	$\left[-x, \frac{7}{8}, \frac{7}{8}\right]$	`[5,14]`
6	$\left[-x, \frac{3}{8}, \frac{3}{8}\right]$	`[6,13]`
7	$\left[x + \frac{1}{4}, \frac{7}{8}, \frac{3}{8}\right]$	`[7,16]`
8	$\left[x + \frac{1}{4}, \frac{3}{8}, \frac{7}{8}\right]$	`[8,15]`
9	$\left[x + \frac{1}{2}, \frac{1}{8}, \frac{5}{8}\right]$	`[17,26]`
10	$\left[x + \frac{1}{2}, \frac{5}{8}, \frac{1}{8}\right]$	`[18,25]`
11	$\left[\frac{1}{4} - x, \frac{1}{8}, \frac{1}{8}\right]$	`[19,28]`
12	$\left[\frac{1}{4} - x, \frac{5}{8}, \frac{5}{8}\right]$	`[20,27]`
13	$\left[\frac{1}{2} - x, \frac{7}{8}, \frac{3}{8}\right]$	`[21,30]`
14	$\left[\frac{1}{2} - x, \frac{3}{8}, \frac{7}{8}\right]$	`[22,29]`
15	$\left[x + \frac{3}{4}, \frac{7}{8}, \frac{7}{8}\right]$	`[23,32]`
16	$\left[x + \frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	`[24,31]`

Table 6: Wyckoff site: `16f`, site symmetry: `.2.`

No.	position	mapping
1	$\left[\frac{1}{8}, y, \frac{1}{8}\right]$	`[1,19]`
2	$\left[\frac{1}{8}, \frac{3}{4} - y, \frac{5}{8}\right]$	`[2,20]`
3	$\left[\frac{5}{8}, y, \frac{5}{8}\right]$	`[3,17]`
4	$\left[\frac{5}{8}, \frac{3}{4} - y, \frac{1}{8}\right]$	`[4,18]`

continued ...

Table 6

No.	position	mapping
5	$\left[\frac{7}{8}, -y, \frac{7}{8}\right]$	[5,23]
6	$\left[\frac{7}{8}, y+\frac{1}{4}, \frac{3}{8}\right]$	[6,24]
7	$\left[\frac{3}{8}, -y, \frac{3}{8}\right]$	[7,21]
8	$\left[\frac{3}{8}, y+\frac{1}{4}, \frac{7}{8}\right]$	[8,22]
9	$\left[\frac{1}{8}, y+\frac{1}{2}, \frac{5}{8}\right]$	[9,27]
10	$\left[\frac{1}{8}, \frac{1}{4}-y, \frac{1}{8}\right]$	[10,28]
11	$\left[\frac{5}{8}, y+\frac{1}{2}, \frac{1}{8}\right]$	[11,25]
12	$\left[\frac{5}{8}, \frac{1}{4}-y, \frac{5}{8}\right]$	[12,26]
13	$\left[\frac{7}{8}, \frac{1}{2}-y, \frac{3}{8}\right]$	[13,31]
14	$\left[\frac{7}{8}, y+\frac{3}{4}, \frac{7}{8}\right]$	[14,32]
15	$\left[\frac{3}{8}, \frac{1}{2}-y, \frac{7}{8}\right]$	[15,29]
16	$\left[\frac{3}{8}, y+\frac{3}{4}, \frac{3}{8}\right]$	[16,30]

Table 7: Wyckoff site: 16g, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{8}, \frac{1}{8}, z\right]$	[1,28]
2	$\left[\frac{1}{8}, \frac{5}{8}, \frac{3}{4}-z\right]$	[2,27]
3	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{4}-z\right]$	[3,26]
4	$\left[\frac{5}{8}, \frac{5}{8}, z\right]$	[4,25]
5	$\left[\frac{7}{8}, \frac{7}{8}, -z\right]$	[5,32]
6	$\left[\frac{7}{8}, \frac{3}{8}, z+\frac{1}{4}\right]$	[6,31]
7	$\left[\frac{3}{8}, \frac{7}{8}, z+\frac{1}{4}\right]$	[7,30]
8	$\left[\frac{3}{8}, \frac{3}{8}, -z\right]$	[8,29]
9	$\left[\frac{1}{8}, \frac{5}{8}, z+\frac{1}{2}\right]$	[9,20]
10	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{4}-z\right]$	[10,19]
11	$\left[\frac{5}{8}, \frac{5}{8}, \frac{1}{4}-z\right]$	[11,18]
12	$\left[\frac{5}{8}, \frac{1}{8}, z+\frac{1}{2}\right]$	[12,17]
13	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{2}-z\right]$	[13,24]
14	$\left[\frac{7}{8}, \frac{7}{8}, z+\frac{3}{4}\right]$	[14,23]
15	$\left[\frac{3}{8}, \frac{3}{8}, z+\frac{3}{4}\right]$	[15,22]
16	$\left[\frac{3}{8}, \frac{7}{8}, \frac{1}{2}-z\right]$	[16,21]

Table 8: Wyckoff site: 32h, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[x, \frac{3}{4}-y, \frac{3}{4}-z\right]$	[2]
3	$\left[\frac{3}{4}-x, y, \frac{3}{4}-z\right]$	[3]
4	$\left[\frac{3}{4}-x, \frac{3}{4}-y, z\right]$	[4]
5	$[-x, -y, -z]$	[5]
6	$\left[-x, y+\frac{1}{4}, z+\frac{1}{4}\right]$	[6]
7	$\left[x+\frac{1}{4}, -y, z+\frac{1}{4}\right]$	[7]
8	$\left[x+\frac{1}{4}, y+\frac{1}{4}, -z\right]$	[8]

continued ...

Table 8

No.	position	mapping
9	$\left[x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9]
10	$\left[x,\, \frac{1}{4}-y,\, \frac{1}{4}-z\right]$	[10]
11	$\left[\frac{3}{4}-x,\, y+\frac{1}{2},\, \frac{1}{4}-z\right]$	[11]
12	$\left[\frac{3}{4}-x,\, \frac{1}{4}-y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[13]
14	$\left[-x,\, y+\frac{3}{4},\, z+\frac{3}{4}\right]$	[14]
15	$\left[x+\frac{1}{4},\, \frac{1}{2}-y,\, z+\frac{3}{4}\right]$	[15]
16	$\left[x+\frac{1}{4},\, y+\frac{3}{4},\, \frac{1}{2}-z\right]$	[16]
17	$\left[x+\frac{1}{2},\, y,\, z+\frac{1}{2}\right]$	[17]
18	$\left[x+\frac{1}{2},\, \frac{3}{4}-y,\, \frac{1}{4}-z\right]$	[18]
19	$\left[\frac{1}{4}-x,\, y,\, \frac{1}{4}-z\right]$	[19]
20	$\left[\frac{1}{4}-x,\, \frac{3}{4}-y,\, z+\frac{1}{2}\right]$	[20]
21	$\left[\frac{1}{2}-x,\, -y,\, \frac{1}{2}-z\right]$	[21]
22	$\left[\frac{1}{2}-x,\, y+\frac{1}{4},\, z+\frac{3}{4}\right]$	[22]
23	$\left[x+\frac{3}{4},\, -y,\, z+\frac{3}{4}\right]$	[23]
24	$\left[x+\frac{3}{4},\, y+\frac{1}{4},\, \frac{1}{2}-z\right]$	[24]
25	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[25]
26	$\left[x+\frac{1}{2},\, \frac{1}{4}-y,\, \frac{3}{4}-z\right]$	[26]
27	$\left[\frac{1}{4}-x,\, y+\frac{1}{2},\, \frac{3}{4}-z\right]$	[27]
28	$\left[\frac{1}{4}-x,\, \frac{1}{4}-y,\, z\right]$	[28]
29	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[29]
30	$\left[\frac{1}{2}-x,\, y+\frac{3}{4},\, z+\frac{1}{4}\right]$	[30]
31	$\left[x+\frac{3}{4},\, \frac{1}{2}-y,\, z+\frac{1}{4}\right]$	[31]
32	$\left[x+\frac{3}{4},\, y+\frac{3}{4},\, -z\right]$	[32]